



SENTIENT TOKEN AND ECONOMIC OPTIMIZATION FOR AGENTS

Make agent optimization falsifiable.

An open operation-control and evidence layer that binds what an agent was asked to do, what actually ran, what it cost, and whether an external verifier accepted the result.

PUBLIC-GOODS REQUEST

\$150,000

Nine months. Three go/no-go tranches. No equity. Open methods, code, evidence, and negative results.

808

public GitHub stars

12 / 12

verified under both policies

-38.7%

hybrid comparison cost

OPEN

MIT code and public proof

PROBLEM

The model call is not the economic unit.

Real cost and failure accumulate across the entire operation.

- Decomposition, parallelism, retries, tools, context, recovery, and verification can overwhelm a cheap model choice.
- A requested model can differ from the provider path that actually executes.
- Unknown local, subscription, setup, or human cost is often collapsed into zero.
- Model prose or stack status can be promoted to completion without an external artifact check.

01 Declared plan

Models, tools, topology, and limits

02 Observed execution

Calls, attempts, artifacts, and cost lenses

03 External verdict

Deterministic verifier outside the route

04 Signed evidence

Offline-reconstructable receipt chain

UNKNOWN STAYS UNKNOWN



WORKING SUBSTRATE

Built before the grant.

Citadel starts with one command, then exposes stronger control only when the operation needs it.

- /do

Beginner path

Natural-language routing, verification, and a durable next action.
- PLAN

Portable contract

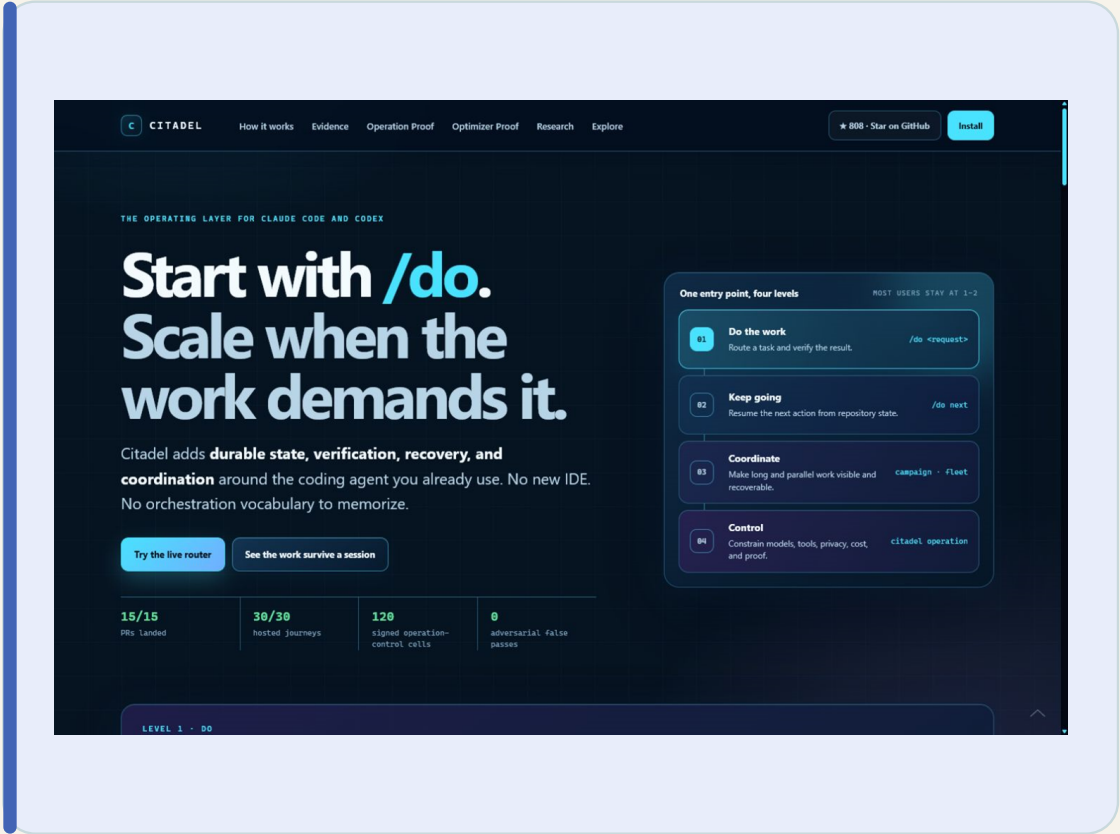
Models, tools, topology, retries, privacy, time, cost, and stop rules.
- RUN

Runtime coverage

Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- PROOF

Evidence plane

External artifact verdicts, failure preservation, signed receipts, and offline replay.



PROSPECTIVE CLAUDE PLUS LOCAL HYBRID V2

Same verified outcomes. Seven fewer Claude calls.

A calibrated support envelope routed bounded operations to local 3B, recovered one rejected artifact, and preserved every strong-baseline completion.

12 / 12

Citadel verified tasks

12 / 12

always-Claude verified tasks

12 -> 5

Claude calls

-38.7%

comparison cost

WHAT THE CONTROLLER DID

- Eight local 3B attempts inside four frozen supported operation shapes.
- One verifier rejection recovered with Claude; no failed output counted.
- Every identity, path, false-pass, cost, quality, and integrity gate passed.

EXACT CLAIM BOUNDARY

Twelve new author-selected synthetic tasks, one model pair, and one workstation. Comparison USD is not the allocated Pro subscription bill; whole-system cost remains unknown.

BOUNDED GATE PASSED



BUILT BEFORE FUNDING

One maintainer already shipped the substrate.

Citadel is a public system with continuing repository pull, not a grant-funded mockup.

134

days public

496 / 520

main commits attributed to Seth

808 / 79

stars / forks

524

unique cloners in 14 days

WORKING SUBSTRATE

- Claude Code, Codex, Ollama, and a pinned Sentient ROMA binding.
- 49 workflows and 35 hooks across 29 lifecycle events.
- Windows, Linux, and macOS verification on every release path.

CLAIM DISCIPLINE

Failed frozen policies remain public. Their baseline and escalation defects became a narrower support envelope that later passed every held-out gate.

FUNDING SCALES A WORKING BASE



NINE-MONTH RESEARCH AND ENGINEERING PROGRAM

Funding buys external validity.

Turn one passed bounded result into an independently valid public optimizer.

01	Representative workload	60+ artifact-producing operations and complete cost boundary	\$30k
02	Adapter SDK	Sentient ROMA plus two additional open stacks	\$28k
03	Expected-value controller	Price verification, escalation, recovery, and failure	\$35k
04	Prospective evaluation	Frozen multi-stack policy and statistical contract	\$32k
05	Public proof and release	Hosted reconstruction, /do onboarding, support window, and maintenance	\$25k



NO-GO RULES ARE PART OF THE PRODUCT

Capped spend. Hard outcome gates.

COST BASIS

Maintainer labor	\$99,000
Evaluation compute and tools	\$27,000
Hosted verification and retention	\$7,000
Reference hardware and measurement	\$7,000
Compatibility and accessibility costs	\$2,000
Bounded contingency	\$8,000

Total **\$150,000**

PRIMARY PROSPECTIVE GATE

> = 80%

absolute verified completion

> = 95%

of valid frontier rate

> = 30%

lower measured end-to-end cost

ZERO

false passes and integrity failures

UNKNOWN COST CANNOT COUNT AS SAVINGS

DIRECT FIT WITH A NAMED REQUEST FOR PRODUCTS

An open optimizer that can prove when it is wrong.

WHY FUND CITADEL

- Directly targets Sentient's Token and Economic Optimization request.
- Already binds a pinned Sentient ROMA stack without replacing its planner.
- A passed 38.7% bounded result lowers substrate risk and exposes the remaining generalization risk.

WHAT STAYS PUBLIC

- MIT code, adapters, contracts, methods, verifiers, cells, and reports.
- No mandatory Citadel account, token, hosted evidence endpoint, or telemetry resale.
- Users keep project state, operations, receipts, and the right to run locally.

Seth Gammon - solo builder and maintainer

Evaluator: github.com/SethGammon/Citadel/blob/main/docs/grants/EVALUATOR_START_HERE.md

OPEN. VERIFIABLE. FALSIFIABLE.